RISHIKESH KUMAR

AI Research Engineer | Voice AI, LLMs & Agentic Systems | Co-founder & CTO

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PROFESSIONAL SUMMARY

AI research engineer and technical founder with 10+ years in software development, including 8+ years in AI and generative systems. Specializes in real-time voice agents, multilingual speech synthesis, LLM architectures, RAG, multi-agent systems, and efficient inference. Co-founder and CTO of Dubpro.ai; most recently completed a six-month AI consulting engagement with MyClone.is. Builds production AI systems and research-driven model implementations, from data pipelines and training to GPU optimization and deployment.

TECHNICAL SKILLS

AI / Model Development: PyTorch, Hugging Face, Transformers, LLM architectures, hybrid/linear attention, state space models (Mamba), SFT, RLHF, RLVR, GRPO, distillation

Voice / Speech AI: Text-to-Speech (TTS), voice cloning, STT, vocoders, neural audio codecs, real-time audio, LiveKit, Pipecat, Twilio

Agents / Retrieval: Model Context Protocol (MCP), multi-agent orchestration, tools and skills, RAG, LlamaIndex, LangChain, pgvector, vector search, persistent memory

Inference / Performance: Triton, CUDA, Flash Attention, vLLM, KV cache, quantization, latency and GPU memory optimization

Backend / Cloud: Python, FastAPI, PostgreSQL, Redis, SQL, Docker, AWS, GCP; additional experience with C++, Rust, Java

PROFESSIONAL EXPERIENCE

AI Consultant / Developer | MyClone.is (formerly Rappo) | 6-month engagement | Remote (San Francisco-based)

- Developed backend features for a real-time voice AI digital persona platform that reproduces a user's voice, conversational style, knowledge, and communication patterns from text and voice data.
- Built low-latency conversational voice workflows with LiveKit, streaming STT/TTS, LLM orchestration, agent tools, and real-time session management.
- Implemented persona-aware retrieval and memory using LlamaIndex, PostgreSQL/pgvector, hybrid RAG, document ingestion, and source-level access filters.
- Migrated embeddings from 1,536 to 512 dimensions; reduced vector storage about 3x, improved retrieval speed 2–2.5x, and reduced voice-chat latency approximately 15–20%.

Open source: github.com/myclone-dev/myclone

Co-founder & CTO | Dubpro.ai (DeepSync Technologies) | Jan 2019 - Present | Noida, India

- Co-founded and led the development of AI dubbing, multilingual text-to-speech, voice cloning, speech transformation, and generative audio systems.
- Architected agentic RAG-enabled dubbing workflows with human-in-the-loop review, increasing dubbing speed by approximately 2x.
- Developed an in-house dubbing harness and tools, including an LLM-as-judge quality verifier, to automate dubbing and quality analysis, reduce reliance on human review, and save approximately 40% in time and 25% in costs.

- Developed multilingual foundation-model capabilities and applied RLHF-based post-training for translation-specific quality and alignment.

EARLIER EXPERIENCE

Data Science / Software Development Engineer | Boutiqaat | Oct 2018 - Jan 2019 | Gurugram, India

- Built text-and-image product matching, hybrid recommendation, and automated product categorization tools for large commercial catalogs.
- Developed clustering, tagging, and product-labeling algorithms to support e-commerce data operations.

Data Scientist | Humonics Global | Dec 2017 - Oct 2018 | Gurugram, India

- Built automated question-answer generation systems for reading comprehension and an LSTM/WaveNet-based TTS engine for conversational applications.
- Worked on face recognition and liveness detection solutions.

Software Engineer | Nucleus Software | Jun 2015 - Dec 2017 | India

- Developed banking transaction analytics and loan-default prediction systems.
- Designed and implemented financial supply-chain management software using MVC architecture.

SELECTED OPEN-SOURCE PROJECTS

Nexus Code Agent - CLI-first AI coding agent with MCP tools, skills, multi-agent orchestration, session/context state, persistent memory, and sandboxed agent workflows. github.com/rishikksh20/nexus-code-agent

Voice AI Agent - Real-time AI assistant supporting inbound/outbound Twilio calls, browser-based WebRTC, and chat with Pipecat-based STT-LLM-TTS pipelines. github.com/rishikksh20/voice-ai-agent-basic

MyClone - Open-source real-time digital persona and voice AI backend using LiveKit, LlamaIndex, FastAPI, PostgreSQL/pgvector, RAG, and agent orchestration. github.com/myclone-dev/myclone

SELECTED RESEARCH & ARCHITECTURE IMPLEMENTATIONS

LLMs and efficient inference - PyTorch implementations of LFM2.5 2.6B (hybrid convolution/attention and Triton inference), Mamba-3, Qwen3.5 0.8B, and Gemma 3 270M.

Speech and generative audio - Implementations and experiments covering Voxtral Codec, NU-Wave, iSTFTNet, VocGAN, and neural audio codec architectures.

Additional model research - Explorations of ViViT, CvT, LocalViT, FNet, CoaT, SiT, and related deep-learning architectures.

Code portfolio: github.com/rishikksh20

EDUCATION

B.Tech, Electronics and Instrumentation Engineering | National Institute of Technology Silchar | 2011 - 2015

LANGUAGES

Hindi (native) | English (proficient)